

Data Sources in Population Health

Tools and Resources for Understanding Community Health

POPH 001: Introduction to Population Health

Overview

- County Health Rankings & Roadmaps (CHR&R)
- Behavioral Risk Factor Surveillance System (BRFSS)
- Social Vulnerability Index (SVI)
- USDA Food Atlas
- Institute for Health Metrics & Evaluation (IHME)
- World Health Organization (WHO)
- Centers for Disease Control (CDC)
- CMS Data
- State & Local Health Departments
- Social Explorer

County Health Rankings & Roadmaps

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What is CHR&R?

Mission

- University of Wisconsin Population Health Institute program
- Highlights health differences within and across communities
- Supported by Robert Wood Johnson Foundation

Vision

- All people and places have what they need to thrive
- Shared understanding of health and equity
- Community power to improve health for all

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CHR&R Snapshot

National Coverage

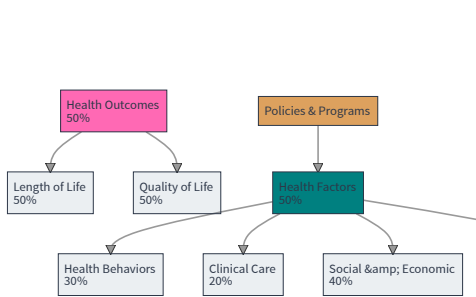
Provides health data for **nearly every county** in the United States

Key metrics tracked:

- Length of life (mortality)
- Quality of life (morbidity)
- Health behaviors
- Clinical care
- Social and economic factors
- Physical environment

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CHR&R Model of Health



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CHR&R Resources

Data & Reports

- Annual national report
- County-level data snapshots
- Trending and comparisons

Evidence Base

- What Works for Health database
- Evidence-informed policies
- Programs and system changes

Tools & Training

- Narratives for Health trainings
- Live and archived webinars
- Podcasts (*In Solidarity*)
- Regular blog updates

Using CHR&R Data

Website: www.countyhealthrankings.org

1. Select your state
2. Compare counties within state
3. Examine specific measures
4. Download raw data for analysis

Behavioral Risk Factor Surveillance System (BRFSS)

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What is BRFSS?

The nation's premier health survey system

- Established in 1984
- CDC-coordinated
- State-based telephone surveys
- Over 400,000 adult interviews annually

Coverage

All 50 states, DC, and 3 US territories

BRFSS Data Collection

Survey method:

- Random-digit-dialed telephone interviews
- Cell phone and landline
- Standardized questionnaire
- State-specific modules available

Frequency:

- Continuous, year-round data collection
- Annual data releases
- Trend analysis possible

BRFSS Health Topics

Chronic conditions

- Diabetes
- Heart disease
- Asthma
- Cancer screening

Health behaviors

- Tobacco use
- Alcohol consumption
- Physical activity
- Nutrition

Preventive services

- Vaccination status
- Health screenings
- Routine checkups

Emerging issues

- Mental health
- COVID-19 impact
- Social determinants

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Accessing BRFSS Data

Website: www.cdc.gov/brfss

- Interactive data tools (WEAT)
- State-level prevalence tables
- County-level estimates (limited)
- Raw microdata available
- SAS, SPSS, R-friendly formats

World Health Organization (WHO)

What is WHO?

Overview

- United Nations health agency
- Global health leadership
- 194 member states
- Headquarters in Geneva, Switzerland

Mission

- Promote health worldwide
- Keep the world safe
- Serve vulnerable populations

Key responsibilities:

- Set health standards and guidelines
- Coordinate global health responses
- Monitor health trends
- Support country health systems
- Conduct health research

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WHO Data Resources

Website: www.who.int/data/#dashboards

Global Health Observatory (GHO)

- Mortality and morbidity
- Health systems data
- Risk factors
- Infectious disease surveillance
- Non-communicable diseases
- Health equity indicators

Data portals:

- World Health Statistics reports
- Disease outbreak dashboards
- SDG health targets tracking
- Country health profiles
- Regional databases

When to Use WHO Data

Best for:

- International health comparisons
- Global disease burden estimates
- Low- and middle-income country data
- Standardized global metrics
- Cross-country policy analysis
- International health regulations

Example uses:

- “How does US life expectancy compare to other wealthy nations?”
- “What is malaria burden in sub-Saharan Africa?”
- “Which countries meet vaccination targets?”
- “Global trends in maternal mortality”

Centers for Disease Control (CDC)

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What is CDC?

Overview

- US national public health agency
- Part of Department of Health & Human Services
- Based in Atlanta, Georgia
- Founded in 1946

Mission

- Protect Americans from health threats
- Disease prevention and control
- Health promotion
- Emergency preparedness

Major centers:

- National Center for Health Statistics (NCHS)
- National Center for Chronic Disease Prevention
- National Center for Immunization
- National Center for Injury Prevention
- Center for Surveillance, Epidemiology, and Laboratory Services

CDC Data Systems

Major surveillance systems:

- National Vital Statistics System
- National Health Interview Survey
- National Health and Nutrition Examination Survey (NHANES)
- Youth Risk Behavior Survey
- Healthcare-associated Infections tracking
- Notifiable Disease Surveillance

Data portals:

- CDC WONDER (mortality, births, cancer)
- CDC Data & Statistics homepage
- FluView (influenza surveillance)
- COVID Data Tracker
- National Environmental Public Health Tracking

Using CDC Data

CDC WONDER:

- Detailed mortality data by cause
- Natality (birth) data
- Cancer incidence
- Vaccination coverage
- Query by geography, demographics

Best for:

- US-specific health data
- Cause-specific mortality
- Real-time outbreak tracking
- Detailed demographic breakdowns
- State and county comparisons

Social Vulnerability Index (SVI)

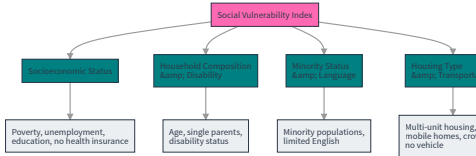
What is SVI?

CDC/ATSDR tool
Helps identify communities that need support before, during, and after hazardous events
Based on US Census data

Purpose

Measure community resilience to stressors

SVI Four Themes



Using SVI

Website: www.atsdr.cdc.gov/placeandhealth/svi

Applications:

- Emergency preparedness planning
- Disaster response resource allocation
- Health equity analysis
- Identifying vulnerable populations
- Grant targeting

Geography:

- Census tract level
- County level
- State level

USDA Food Access Research Atlas

Food Access Research Atlas

Purpose

- Maps food access indicators
- Identifies food deserts
- Shows low-income areas with limited access

Data sources:

- Census demographics
- Store locations
- Vehicle availability
- Distance to stores

Food Desert Criteria

Definition

Low-income census tract where significant share of residents is far from a supermarket

Distance thresholds: - Urban: > 1 mile from supermarket - Rural: > 10 miles from supermarket

Income criteria: - Poverty rate ≥ 20%, OR - Median family income ≤ 80% of area median

Using the Food Atlas

Website: www.ers.usda.gov/foodatlas

Key indicators:

- Low access populations
- Vehicle access
- SNAP participation
- Store availability
- Child and senior populations affected

Institute for Health Metrics and Evaluation (IHME)

What is IHME?

University of Washington
research center

- Global health statistics
- Disease burden research
- Comparative risk assessment
- Health system performance

Known for

Global Burden of
Disease (GBD)
Study

IHME Data Products

Major projects:

- **Global Burden of Disease (GBD)** - Comprehensive disease, injury, and risk factor data
- **GBD Compare** - Interactive visualization tool
- **US County-level estimates** - Mortality, life expectancy, risk factors
- **COVID-19 projections** - Pandemic modeling
- **Health inequality analysis**

Accessing IHME Data

Website: www.healthdata.org

Tools:

- GBD Results Tool - query any GBD metric
- GBD Compare - visualize multiple metrics
- Data visualizations - pre-made graphics
- Publications - peer-reviewed research

Geographic levels:

- Global, national, state, county

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WHO & CDC Data

World Health Organization (WHO)

What is WHO?

- United Nations health agency
- Global health leadership
- 194 member states
- Headquarters in Geneva

Key data resources:

- Global Health Observatory (GHO)
- World Health Statistics
- Disease outbreak tracking

Data coverage:

- Mortality and morbidity
- Health systems and services
- Risk factors and NCDs
- Infectious disease surveillance
- Health equity indicators
- SDG health targets

Website: www.who.int/data

CDC

What is CDC?

- US national public health agency
- Disease prevention and control
- Health promotion
- Emergency preparedness

Major data systems:

- National Vital Statistics System
- National Health Interview Survey
- Healthcare-associated infections

Data portals:

- CDC WONDER - mortality, natality, cancer
- CDC Data & Statistics portal
- National Center for Health Statistics
- FluView - influenza surveillance

Website: www.cdc.gov/datastatistics

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WHO vs CDC: When to Use Each

Use WHO data for:

- International comparisons
- Global disease burden
- Low- and middle-income countries
- International health regulations
- Cross-country health system comparisons
- Standardized global metrics

Use CDC data for:

- US-specific health issues
- Detailed US mortality data
- US surveillance systems
- State and local US data
- US health behaviors and risk factors
- Timely US outbreak information

CMS Data

Centers for Medicare & Medicaid Services

What is CMS?

- Federal agency within HHS
- Administers Medicare, Medicaid, CHIP
- Massive healthcare claims database

Data utility:

- Healthcare utilization
- Costs and spending
- Quality measures
- Provider performance

Key CMS Data Sources

Data.CMS.gov offers:

- Medicare enrollment and claims
- Provider utilization patterns
- Hospital quality measures
- Nursing home quality
- Physician compare data
- Drug spending information
- State Medicaid data

CMS Geographic Data

Available geographies:

- National
- State
- County
- ZIP code
- Hospital referral region (HRR)
- Hospital service area (HSA)

Privacy protection

Small cell sizes suppressed to protect beneficiary privacy

State & Local Health Departments

Pennsylvania Department of Health

State-level data:

- Vital statistics
- Communicable diseases
- Environmental health
- Maternal and child health
- Chronic disease surveillance

Local health departments:

- County-specific data
- Community health assessments
- Local program evaluation
- Real-time surveillance

Why Use State/Local Data?

Advantages:

- More timely than national sources
- Greater geographic detail
- Local context and interpretation
- Connection to local programs
- Community engagement opportunities
- Non-aggregated when possible

Pennsylvania Health Data

Website: www.health.pa.gov
Key resources:

- PA Health Statistics
- Enterprise Data Dissemination Informatics Exchange (EDDIE)
- Community Health Needs Assessments
- Vital statistics reports
- Outbreak reports

Social Explorer

What is Social Explorer?

Web-based demographics platform

- Easy-to-use interface
- Census data visualization
- Historical data access
- Map creation tools

Primary source

US Census Bureau data

Social Explorer Features

Data available:

- Decennial Census (1790-2020)
- American Community Survey
- Religious congregations
- Carbon emissions
- American Indian data

Using Social Explorer

Typical workflow:

- 1. Select data source and year
- 2. Choose geographic level
- 3. Select variables of interest
- 4. Create map or table
- 5. Annotate and customize
- 6. Export for presentations/reports

Access: Many universities provide subscriptions

Putting It All Together

Matching Questions to Data Sources

Research Question	Best Data Source
How does my county rank on health outcomes?	CHR&R
What % of adults smoke in my state?	BRFSS
Which neighborhoods are most vulnerable?	SVI
Where are food deserts located?	Food Atlas
What is disease burden in my county?	IHME
Compare US health to other countries?	WHO
US mortality rates by cause?	CDC WONDER
Medicare spending patterns?	CMS
Detailed demographics of a ZIP code?	Social Explorer

Next Steps

Practice using these tools:

- 1. Explore your home county on CHR&R
- 2. Look up health behaviors in BRFSS
- 3. Check SVI for your community
- 4. Map food access in your area

Remember: Different sources provide complementary perspectives

Sources

Resources mentioned today:

- County Health Rankings: www.countyhealthrankings.org
 - BRFSS: www.cdc.gov/brfss
 - SVI: www.atsdr.cdc.gov/placeandhealth/svi
 - Food Atlas: www.ers.usda.gov/foodatlas
 - IHME: www.healthdata.org
- WHO: www.who.int/data
 - CDC: www.cdc.gov/datastatistics
 - CMS: data.cms.gov
 - PA Health: www.health.pa.gov
 - Social Explorer: www.socialexplorer.com

Error